

Youth Drug Use Prediction Using Decision Trees - Script

Slide 1: Title Slide

Hello everyone, my name is Mahek Patel, and today I will be presenting my practical homework project on “Youth Drug Use Prediction”, based on the 2023 NSDUH dataset. In this project, I explored youth behavior regarding alcohol, marijuana, and cigarette use by building three different machine learning models: binary classification, multiclass classification, and regression.

Slide 2: Introduction

The primary goal of this project is to predict and understand youth drug use patterns. The target behaviors are alcohol, marijuana, and cigarette use among individuals under the age of 18. To analyze this, I applied interpretable decision tree-based models, which help identify important features and risk patterns. This kind of analysis is crucial because early identification of such risk factors can guide policy interventions and educational programs aimed at prevention.

Slide 3: Theoretical Background

Decision trees are supervised learning models that split data into smaller groups based on feature thresholds. They're intuitive and easy to interpret, which makes them suitable for public health-related decision-making. Each split is determined by metrics such as Gini impurity or entropy, aiming to increase homogeneity in each node. However, trees can overfit easily, so parameters like `max_depth` are used to prune them. In this project, I've evaluated the models using metrics like accuracy, F1-score for classification, and MSE and R^2 for regression.

Slide 4: Methodology

The dataset used is the cleaned youth version of NSDUH 2023, which includes demographic, behavioral, and environmental data. I used Python libraries such as `pandas` and `scikit-learn` for cleaning, visualization, and modeling. Initial preprocessing involved removing duplicates and dropping columns with more than 10% missing values. I recoded categorical variables—for instance, gender and race—and created a new binary target called `CIG_USED` to indicate cigarette usage. Some of the main features I selected for modeling include `IRCIGFM`, `IRMJFM`, `IRALCFM`, `INCOME`, `SCHFELT`, `IRSEX`, and `NEWRACE2`.

Slide 5: Exploratory Data Analysis

To understand feature relationships, I generated a correlation matrix. Some key correlated features included emotional stress (`SCHFELT`) and peer influence

variables such as FRDMEVR2 and FRDPCIG2. This exploratory analysis helped guide which variables were included in the models and confirmed that emotional and social factors play a role in substance use.

Slide 6: Model 1 – Binary Classification

The first model predicts whether a youth has ever used cigarettes, using the new target variable CIG_USED. I used a decision tree classifier with a max depth of 5. The model's most significant split was based on emotional health, which indicates that stress plays a major role in substance use behavior. Other important predictors included gender, race, and the number of days the youth smoked. Evaluation was done using a confusion matrix and ROC curve, and the model performed well on training data but showed signs of overfitting on testing.

Slide 7: Model 2 – Multi-Class Classification

In this model, the goal was to classify marijuana use frequency (IRMJFM) into categories like seldom, moderate, and frequent. I used a decision tree classifier here as well. Emotional stress again came up as the most informative feature, followed by alcohol frequency and gender. This model helps us segment risk groups based on frequency of use and can guide targeted interventions.

Slide 8: Model 3 – Regression

The third model was a regression tree to predict the number of days alcohol was consumed in the past year (ALCYDAYS). I also trained a linear regression model and a random forest regressor for comparison. The regression model showed that emotional stress and prior marijuana use were the strongest predictors. The best performance came from the linear model, with an MSE of 0.0664 and an R^2 score of 0.978, indicating that it explains nearly all the variation in alcohol use days.

Slide 9: Key Predictors & Interpretation

Across all models, one feature stood out: SCHFELT, the emotional stress variable. This suggests that mental health is a consistent and strong predictor of youth substance use. Other impactful variables included income, gender, and peer influence. The decision trees offered interpretable paths—for instance, a youth with high stress and low parental involvement had a higher likelihood of usage.

Slide 10: Drawbacks & Improvements

There are several areas in this project that need improvement for better implementation:

- **Overfitting** was a major issue, especially with the decision trees and random forests.
- **Class imbalance** affected both the binary and multi-class models.

- I didn't implement **cross-validation** in this iteration, which may limit generalizability.
- Also, I only used a limited number of features out of the 2,890 available.

To improve the overall outcome and an easy understanding, we can also:

- Use SMOTE or class weights to balance the data.
- Apply k-fold cross-validation for stronger validation.
- Explore more variables like parental involvement and mental health indicators to deepen insights.

Slide 11: Ethical Considerations

When dealing with youth behavioral data, it's crucial to communicate results ethically. Predictive models can help identify at-risk groups, but they must be used responsibly. We should avoid reinforcing stereotypes or stigmas. Instead, findings should support supportive interventions, not punitive measures.

Slide 12: Conclusion

In conclusion, decision tree models helped uncover meaningful patterns in youth drug use. Emotional stress was consistently the strongest predictor, followed by race, gender, and income. While the models were interpretable and accurate in some cases, overfitting and limited validation were challenges. With future iterations using ensemble models and broader feature sets, the models can become even more robust and insightful.

Slide 13: Thank You

Thank you for watching this presentation. If you have any questions, I'd be happy to discuss further!